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Diagnosis and Clinical Evaluation of Acute Kidney Injury

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The term *acute renal failure* (ARF) describes the clinical syndrome in which an abrupt (hours to days) decrease in kidney function leads to the accumulation of nitrogenous waste products and, commonly, a reduction in urine output. *Acute kidney injury* (AKI) has become the consensus term for ARF.[1-3] This change in terminology served to standardize a definition for the syndrome, as well as to incorporate knowledge that increases in serum creatinine as small as 0.3 mg/dL (27 μmol/L) are associated with increased morbidity and mortality.[4]

In 2012 the Kidney Disease: Improving Global Outcomes (KDIGO) guideline unified the definition of AKI and included stages of severity (Table 72.1). AKI has since been defined as an increase in serum creatinine of 0.3 mg/dL or greater within 48 hours of observation or 1.5 or more times baseline, which is known or presumed to have occurred within 7 days, or a reduction in urine volume below 0.5 mL/kg/h for 6 hours.[1] Subclassification of AKI based on disease severity as indicated by the level of increase in serum creatinine and reduction in urine output has been adopted as a three-stage classification by KDIGO. Increasing severity of AKI based on creatinine and urine output associates with increased risk for death.[1]

There are many different causes of AKI, and most are identified through clinical investigation. Globally, hypotension and volume contraction or dehydration accounted for 40% of AKI cases in both the hospital and nonhospital setting.[5] In higher-income countries, hypotension and shock were the most common causes of AKI, whereas volume contraction or dehydration predominated in lower-income countries. Nephrotoxic agents were implicated in up to a quarter of AKI events across all countries.[5] However, nephrotoxin administration may account for a larger proportion of hospital-acquired AKIs in older patients.[6,7] Kidney biopsy is rarely performed to establish the cause of AKI. In an observational study of 745 patients with AKI in the intensive care unit (ICU), only 3% (22 patients) underwent kidney biopsy and, as expected, glomerulonephritis (GN) and acute interstitial nephritis (AIN) were each diagnosed in about a third of the biopsy specimens, and about 23% showed only acute tubular necrosis (ATN).[8]

EARLY DETECTION OF ACUTE KIDNEY INJURY

The definition of AKI is based on increases in serum creatinine, yet when used as a marker of kidney function, serum creatinine concentrations have multiple limitations. In addition to a steady-state balance of creatinine production and excretion being required for appropriate estimation of glomerular filtration rate (GFR), serum creatinine concentrations may not rise for subtle declines in GFR and are slow to rise for rapid falls in GFR. Furthermore, the generation of creatinine from muscle is reduced in sepsis-induced AKI, and serum creatinine concentrations may not increase proportionally to GFR decline.[9] Kidney injury will initially remain undetected until serum creatinine concentrations rise (8–48 hours) (Fig. 72.1).[10] Serum and urine biomarkers with potential as early indicators of AKI include tissue inhibitor of metalloproteinase 2 (TIMP-2), insulin-like growth factor–binding protein 7 (IGFBP7), hepcidin, kidney injury molecule 1 (KIM-1), neutrophil gelatinase–associated lipocalin (NGAL), cystatin C, interleukin-18 (IL-18), proenkephalin A, and others. These biomarkers also may allow for improved prognostication and insight into the specific cause of AKI.[11]

Signifying the integration of nontraditional biomarkers in the diagnosis of AKI, the definition of AKI has been expanded to include biomarker status, including a subclinical stage (1S) where biomarkers are positive without significant changes in serum creatinine or urine output.[11] Biomarkers are categorized by predictive type. The strongest predictor of AKI severity and outcome involves a combination of biomarker types and clinical judgment. For example, the combination of biomarkers reflecting kidney function (cystatin C) and kidney damage (NGAL) was superior to serum creatinine in predicting AKI severity in children undergoing cardiopulmonary bypass.[12]

Serum cystatin C is a 13-kDa cysteine protease inhibitor that is freely filtered at the glomerulus and normally reabsorbed by proximal tubule cells and may be more sensitive than serum creatinine concentrations for detecting small reductions in GFR,[13] and thus may have advantages for diagnosing AKI and predicting AKI severity.[11,14]

The combination of urinary levels or IGFBP7 and TIMP outperforms all other biomarkers in the early detection of AKI in critically ill patients.[15] Both proteins are expressed in tubular cells and have autocrine and paracrine effects that lead to G1 cell cycle arrest for short periods. In AKI, preventing or delaying tubular cell division may be a protective response to injury. The risk for stage 2 or 3 AKI correlated well with logarithmic values of [IGFBP7]*[TIMP-2] measured from a urine sample obtained within 12 hours of the diagnosis of AKI,[15] and their combined urinary levels correlated to clinician-adjudicated AKI better than KDIGO criteria.[16] The release of TGFBP7 and TIMP-2 during cell cycle arrest make their urinary levels potentially useful for predicting the development of AKI.[11,17]

Assessing kidney function with a loop diuretic challenge (furosemide stress test) improves prediction of kidney outcomes compared with biomarker measurement. In patients with early AKI in the ICU, urinary responses to intravenous furosemide predicted the need for dialysis better than biomarker measurement alone.[18] Failure to produce more than 200 mL of urine within 2 hours of intravenous furosemide dosed at 1 to 1.5 mg/kg strongly predicted both the need for dialysis and progression to AKI stage 3.[18]

KIM-1, a cell membrane glycoprotein upregulated in injured proximal tubular cells, can act as a nonmyeloid phosphatidylserine receptor that transforms epithelial cells into semiprofessional phagocytes.[19] The ectodomain of this membrane-associated molecule is shed into the urine of injured but not healthy kidneys. KIM-1 messenger ribonucleic acid

TABLE 72.1 Kidney Disease: Improving Global Outcomes Composite Staging of Acute Kidney Injury

Stage	Serum Creatinine	Urine Output
1	1.5–1.9 × baseline *or* ≥0.3 mg/dL (≥29 μmol/L) increase	<0.5 mL/kg/h for 6–12 h
2	2.0–2.9 × baseline	<0.5 mL/kg/h for ≥12 h
3	3.0 × baseline *or* Increase in serum creatinine to ≥4.0 mg/dL (≥352 μmol/L) *or* Initiation of kidney replacement therapy *or* In patients younger than 18 yr, decrease in estimated glomerular filtration rate <35 mL/min/1.73 m^2	<0.3 mL/kg/h for ≥24 h *or* Anuria for ≥12 h

From the Kidney Disease Improving Global Outcomes (KDIGO) Working Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl.* 2012;2(1):1–138.

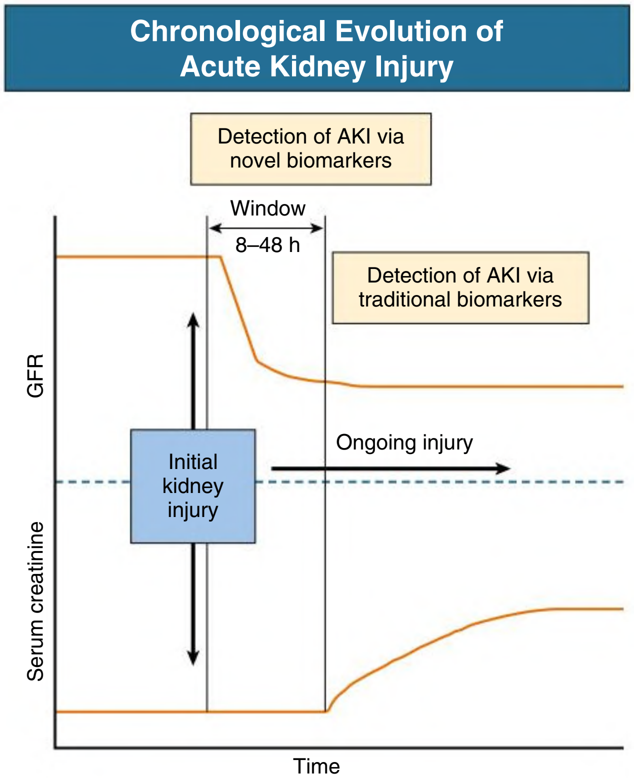

Fig. 72.1 Chronologic evolution of acute kidney injury (AKI). *GFR,* Glomerular filtration rate. (Modified from Ravn B, Rimes-Stigare C, Bell M, et al. Creatinine versus cystatin C based glomerular filtration rate in critically ill patients. *J Crit Care.* 2019;52:136–140.)

(mRNA) levels may rise more than any other gene after kidney injury, and urinary levels are increased specifically with AKI resulting from ischemia or toxin exposure.[20,21] Urine KIM-1 is categorized as a damage biomarker.[11]

NGAL, a protein produced by neutrophils, binds and traffics free iron.[20] It also mediates the tubular response to epidermal growth factor and is thereby involved in the progression of kidney disease.[22] Urinary NGAL levels are increased in the setting of tubular stress or injury, but not in prerenal disease.[23] A large number of studies correlated urinary NGAL levels to early detection of AKI.[10] Plasma and urine levels of NGAL are categorized as a damage biomarker that may be used to diagnose and predict the severity of AKI.[11]

IL-18 is an inflammatory cytokine found in macrophages, monocytes, and proximal tubule cells. Urinary levels are upregulated in kidney ischemic injury in multiple clinical settings.[10] Urinary IL-18 levels are used as a damage biomarker.[11]

Biomarkers have the potential for detecting AKI early, identifying minor kidney injuries that do not raise serum creatinine concentrations (subclinical AKI), monitoring therapeutic benefits of novel treatment interventions, and specifying the cause of AKI. However, AKI frequently occurs unobserved in the community or is present at the time of recognition. Approximately 50% of patients admitted from the emergency department with septic shock have AKI on arrival to the ICU.[24] Even when computer models successfully predict hospital-acquired AKI, early interventions have had little effect on outcomes. The composite outcome of maximum change in serum creatinine level, need for dialysis, or death at 7 days did not change when primary providers were notified of AKI by an electronic alert system.[25] Multicenter implementation of an AKI alert system along with an AKI care bundle and education of health care workers resulted in increased AKI recognition and more frequent assessment of patient fluid status, but the risks of 30-day mortality after AKI (24.5%) and progressive AKI were unchanged.[26] Nevertheless, biomarkers remain an exciting potential tool for diagnosing, identifying a cause of, and predicting outcomes of AKI.

DIAGNOSTIC APPROACH TO ACUTE KIDNEY INJURY

The basic diagnostic approach to patients with AKI is to determine the cause (see Chapter 70). This process should start by excluding or correcting both prerenal and postrenal causes (Fig. 72.2). In hospitalized patients, making a diagnosis often involves selecting the most probable cause among many choices.[27] Assessing daily urine volume can narrow the differential diagnosis, dividing AKI into oliguric (<500 mL/d) and nonoliguric causes. Acute tubular necrosis, the most common cause of hospital-acquired AKI, is oliguric, whereas AIN and glomerular-related AKI are typically nonoliguric. A careful history and physical examination and basic laboratory tests often suffice for diagnosis (Table 72.2).[27]

Acute Kidney Injury Versus Chronic Kidney Disease

It can be difficult to determine whether a patient with kidney failure has AKI or AKI superimposed on chronic kidney disease (CKD). Knowledge of prior serum creatinine concentrations is required to determine the degree of potentially reversible AKI. Ultrasound evidence of small, scarred kidneys is consistent with CKD, but even advanced diabetic nephropathy, amyloidosis, human immunodeficiency virus (HIV) infection–related nephropathy, or polycystic kidney disease can present with normal or increased kidney size. Normocytic anemia, hyperparathyroidism, peripheral neuropathy, and broad waxy casts in the urinary sediment suggest CKD. Patients with CKD are at high risk for developing AKI.[28]

CLINICAL ASSESSMENT

The evaluation of a hospitalized patient with AKI should begin with a complete medical history and review of the hospital records. Once the prior serum creatinine concentration or other evidence of underlying kidney disease is established, the history should be directed toward the events preceding the AKI (see Fig. 72.2). These events

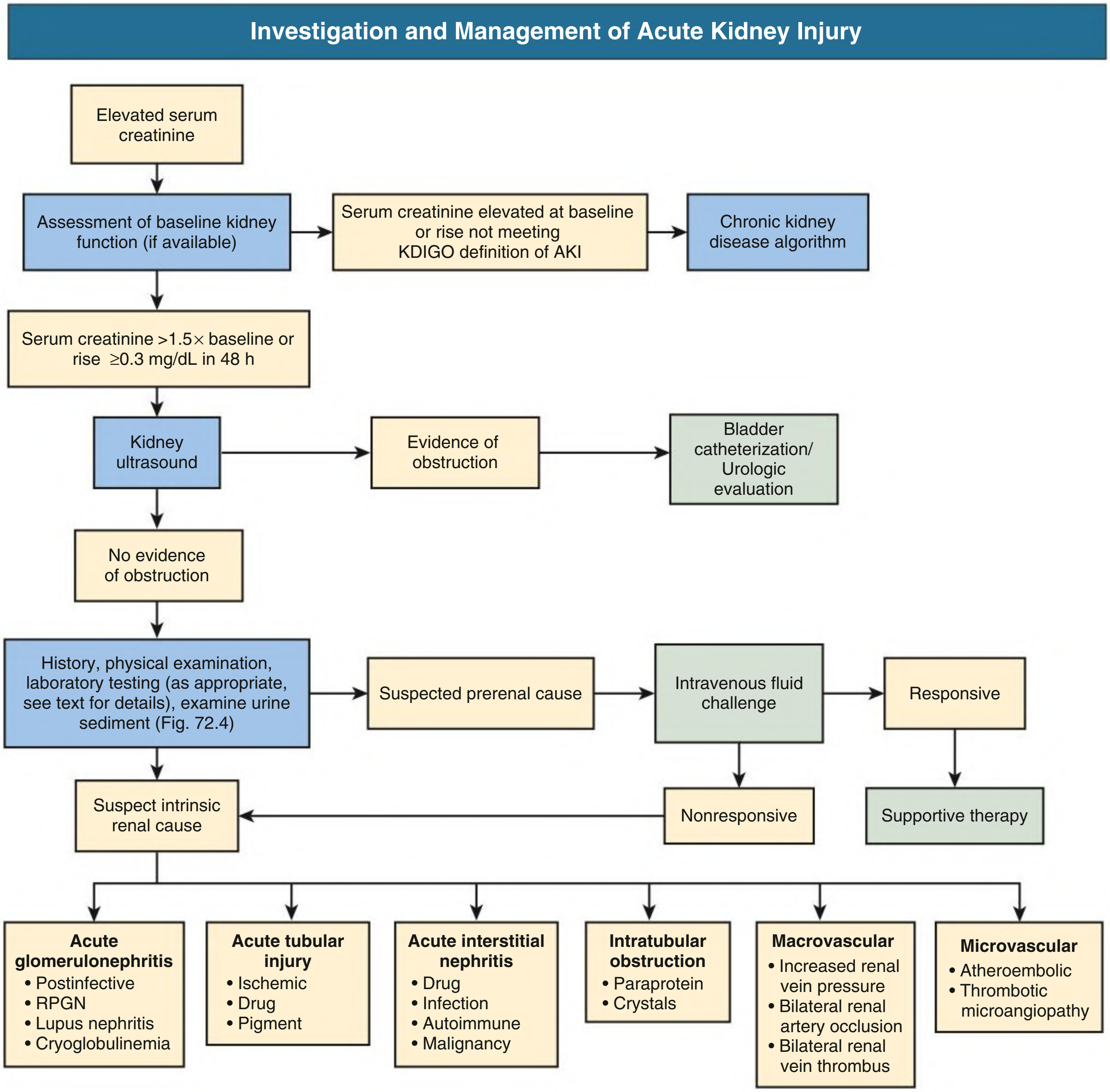

Fig. 72.2 **Investigation and Management of Suspected Acute Kidney Injury (AKI).** The majority of AKI causes can be identified by reviewing the events that preceded the AKI. Evaluating for urinary obstruction early prevents the delay of urologic therapy, if needed; however, not all patients require a kidney ultrasound. If prerenal and postrenal causes have been excluded, the kidneys are normal size, and the diagnosis remains unclear, a kidney biopsy may be indicated. *KDIGO,* Kidney Disease: Improving Global Outcomes; *RPGN,* rapidly progressive glomerulonephritis.

may be part of a systemic disease process (e.g., sepsis, rhabdomyolysis), an inpatient event (e.g., surgery, nephrotoxic agent exposure), or an outpatient event (e.g., medication or drugs, volume contraction from diarrhea or vomiting). Particular attention should be paid to the medication record, looking for use of nonsteroidal antiinflammatory drugs (NSAIDs), renin-angiotensin-aldosterone antagonists, diuretics, antibiotics, proton pump inhibitors, chemotherapy agents (e.g., platin-based therapies and immune check point inhibitors), herbs that might contain aristolochic acid (see Chapter 65), or a synthetic cannabinoid.[29] In Africa and India the ingestion of hair dyes containing paraphenylenediamine may result in AKI. Clues to a postrenal cause (e.g., urinary hesitancy, frequent nocturia, pelvic or flank pain, overflow urinary incontinence, metastatic cancer) should be assessed early.

Physical examination may reveal reduced body weight, marked orthostatic decrease in blood pressure, an increase in pulse rate, and lack of jugular venous distention, all suggesting a reduction in extracellular fluid volume. Patients with prerenal AKI can appear volume overloaded in heart failure, cirrhosis, and nephrotic syndrome when effective arterial blood volume is reduced. In critically ill patients,

TABLE 72.2 Differential Diagnosis by Pathophysiologic Classification of AKI

Cause	Comments
Prerenal	**30%–60% of AKI**
Volume depletion	Renal losses, GI losses, hemorrhage
Decreased cardiac output	Right- or left-sided heart failure, cardiac tamponade
Systemic vasodilation	Sepsis, anaphylaxis, anesthetics
Afferent arteriolar vasoconstriction	NSAIDs, calcineurin inhibitors, radiocontrast, hepatorenal syndrome, hypercalcemia
Efferent arteriolar vasodilation	ACE inhibitors, ARBs
Intrinsic	**~40% of AKI**
Acute tubular injury	
Ischemic	
Nephrotoxic (drug)	Aminoglycosides, lithium, amphotericin, pentamidine, cisplatin, ifosfamide, radiocontrast
Nephrotoxic (pigment)	Rhabdomyolysis, intravascular hemolysis
Acute interstitial nephritis	
Drug induced	Penicillins, cephalosporins, NSAIDs, PPIs, allopurinol, rifampin, sulfonamides, immune checkpoint inhibitors
Infection related	Pyelonephritis, viral nephritides
Autoimmune diseases	Sjögren syndrome, sarcoidosis, SLE
Malignancy	Lymphoma, leukemia
Intratubular obstruction	
Paraprotein	Immunoglobulin light chains
Crystals	Acute phosphate nephropathy, tumor lysis syndrome, ethylene glycol, acyclovir, indinavir, methotrexate
Acute glomerulonephritis	Postinfectious, cryoglobulinemia, RPGN, SLE
Macrovascular	Increased renal vein pressure from increased intraabdominal pressure, bilateral renal vein thrombosis, bilateral renal artery emboli
Microvascular	Atheroembolic disease, HUS, TTP, scleroderma renal crisis, malignant hypertension
Postrenal (Obstruction)	**Approximately 10% of AKI**
Intrinsic	Bilateral ureteral stones, bladder outlet obstruction (prostatic enlargement or blood clot), neurogenic bladder
Extrinsic	Retroperitoneal fibrosis, metastatic cancer

ACE, Angiotensin-converting enzyme; *AKI,* acute kidney injury; *ARBs,* angiotensin receptor blockers; *GI,* gastrointestinal; *HUS,* hemolytic uremic syndrome; *NSAIDs,* nonsteroidal antiinflammatory drugs; *PPIs,* proton pump inhibitors; *RPGN,* rapidly progressive glomerulonephritis; *SLE,* systemic lupus erythematosus; *TTP,* thrombotic thrombocytopenic purpura.

hemodynamic measurement by right heart catheterization may be necessary to differentiate volume overload from noncardiogenic pulmonary infiltrates. Trends in daily intake and output volumes also assist in determining the extracellular fluid volume of the critically ill patient. Monitoring urine output in the ICU is associated with improved detection of AKI, as well as reduced 30-day mortality in patients with AKI.[30]

Examination of the abdomen may reveal a tender, distended bladder in a lower urinary tract obstruction, and, when this is present, sterile postvoid bladder catheterization should be performed. A distended, tense abdominal wall may represent ascites, aggressive intravenous fluid resuscitation, or recent abdominal surgery. Intraabdominal pressure can be measured in the ICU to distinguish AKI from abdominal compartment syndrome, defined as intraabdominal pressure exceeding 20 mm Hg.[31]

Fever, skin rash, and arthralgias may be signs of systemic lupus erythematosus, vasculitis, endocarditis, or drug allergy with AIN. A history of recent aortic catheterization (e.g., cardiac catheterization) and the findings of livedo reticularis or a discolored toe are diagnostic clues for cholesterol or atheromatous emboli. Painless hematuria suggests acute GN or genitourinary malignancy, whereas painful hematuria is more consistent with obstruction.[27]

DIAGNOSTIC EVALUATION

Differentiating the two most common causes of AKI in hospitalized patients, prerenal AKI and ATN, may be difficult when both the effective arterial blood volume and the time course of the kidney injury are unknown.[32] Here the term *acute tubular injury* may more accurately describe the pathology involved in intrinsic AKI from ischemic or toxic insults.[33] Evaluation of urine volume, urinary sediment, and urinary indices (the last is useful only in patients with oliguria) is helpful in making the correct diagnosis (Table 72.3 and Fig. 72.2). Initial laboratory tests include a urinalysis, blood urea nitrogen (BUN), serum sodium, potassium, bicarbonate, and creatinine levels not only for the diagnosis but also for assessment of complications of AKI.[27] The dipstick urinalysis, a low-cost test that provides rapid information about urinary protein and cellular content, is commonly underutilized. Among 3 tertiary care hospitals, a urinalysis was obtained less than half the time in patients diagnosed with AKI.[34]

Results from initial laboratory testing may prompt further evaluation. For example, in the absence of treatment with a sodium-glucose cotransporter-2 inhibitor, glycosuria occurring with a normal plasma glucose level offers evidence for proximal tubular dysfunction. The presence of amino acids and bicarbonate in the urine and/or elevated urinary fractional excretion of phosphate and uric acid can be used to confirm Fanconi syndrome (see Chapter 50).

Ratio of Blood Urea Nitrogen to Creatinine

The ratio of BUN to creatinine in healthy individuals is 10:1 to 15:1 (when both are expressed in mg/dL, or 40:1 to 60:1 when expressed in mmol/L). In prerenal AKI the ratio may exceed 20:1 because of a disproportionate increase in urea reabsorption resulting from elevated serum vasopressin levels. However, BUN-to-creatinine ratio has limited value in differentiating the type of AKI. Upper gastrointestinal tract bleeding, impaired protein anabolism (e.g., systemic corticosteroid administration), increased catabolism (e.g., sepsis), and increased

TABLE 72.3 Clinical and Laboratory Variables in the Differential Diagnosis Between Prerenal and Intrarenal Acute Kidney Injury

	Prerenal	Intrarenal
History	Volume loss from GI, urinary, skin, or blood or reduced EABV (e.g., heart failure, pancreatitis)	Drugs or toxin exposure, hemodynamic change
Clinical presentation	Hypotension or volume depletion	No specific symptoms or signs
Laboratory studies		
BUN/S_{Cr}	>20	<20
Sediment	Normal to few hyaline casts	Muddy brown casts
U_{osm} (mmol/kg)	>500	<350
Proteinuria	None to trace	Mild to moderate
U_{Na} (mmol/L)	<20	>40
FE_{Na} (%)	<1	>1
FE_{Urea} (%)	<35	>35
Novel biomarkers	None	IGFBP7*TIMP-2, KIM-1, cystatin C, NGAL, CYR61, others

BUN, Blood urea nitrogen; *CYR61,* cysteine-rich protein 61; *EABV,* effective arterial blood volume; FE_{Na}, fractional excretion of sodium; FE_{Urea}, fractional excretion of urea; *GI,* gastrointestinal; IGFBP7*TIMP-2, insulin-like growth factor-binding protein 7 and tissue inhibitor of metalloproteinase 2; *KIM-1,* kidney injury molecule 1; *NGAL,* neutrophil gelatinase–associated lipocalin; S_{Cr}, serum creatinine; U_{Na}, urinary sodium; U_{osm}, urinary osmolality.

protein intake all raise BUN levels. Furthermore, diminished urea production from decreased protein intake or underlying liver disease are associated with lower BUN levels. Also, elevations in serum creatinine levels may exceed BUN levels in patients with creatinine release from muscle breakdown, as in rhabdomyolysis.

Urine Volume

In AKI, urine volume directly correlates with residual GFR.[35] Urine volume can indicate the severity of AKI and also provide important diagnostic information. Oliguric AKI (<500 mL/d) is typically associated with worse outcomes than AKI with preserved urine volume, especially with a positive fluid balance in the critical care setting.[36-39] Oliguria commonly occurs in AKI caused by ATN, although it can be seen in early prerenal AKI or AIN. Wide variations in daily urine output suggest obstruction. Anuria (urine output <100 mL/d) suggests obstruction or a bilateral acute vascular catastrophe (renal vein or renal artery occlusion), shock, rarely kidney cortical necrosis, hemolytic uremic syndrome, or anti–glomerular basement membrane (GBM) antibody disease.[27]

Urinalysis and Urine Microscopy

In the setting of AKI, a dipstick urinalysis may reveal microhematuria or proteinuria but must be interpreted in conjunction with more specific tests such as a ratio of spot urinary protein or albumin to creatinine and urine microscopy. Dipstick urinalysis may not detect immunoglobulin free light chains (FLCs) or may be falsely positive for protein in the setting of radiographic contrast or alkaline urine. In conjunction with urine microscopy, the presence of urinary hemoglobin or myoglobin by dipstick can be further differentiated. Urine microscopy has been validated as a diagnostic and prognostic tool in hospitalized patients with AKI.[32] A fresh urine sample is centrifuged, and the sediment is examined for the presence of cells, casts, and crystals (Figs. 72.3 and 72.4) (see also Chapter 4). Urine microscopy in early prerenal AKI is typically normal with occasional hyaline casts. "Muddy brown" granular casts and renal tubular epithelial cells in the urine support ATN-related AKI. In hospitalized patients with AKI, the presence of more than 10 granular casts per low-power field had a positive predictive value of 100% for a final diagnosis of ATN, and a urine sediment score based on granular casts and renal tubular epithelial cells was directly associated with worsening AKI. Thus, urine microscopy is useful for distinguishing ATN-related AKI from prerenal AKI and predicting the severity of AKI.[32]

Findings on urinalysis and urine microscopy (see Chapter 4) may suggest CKD (broad, waxy casts), but, more important, they may be diagnostic clues to a rare cause of AKI. Proliferative GN may be characterized by considerable microhematuria and proteinuria by dipstick and an active urinary sediment consisting of red blood cells (RBCs) and RBC casts. In this setting the history and physical examination findings should be supported by serologic testing and a kidney biopsy, if the kidneys are normal in size. The presence of white blood cells (WBCs) in clumps and casts, in the absence of bacteria, suggests AIN.[27] Renal tubular epithelial cells, granular casts, RBCs and, rarely, even RBC casts can occur in patients with AIN, whereas urinary eosinophils are no longer considered helpful in diagnosing AIN (see Chapter 64).[40] A urine sediment with abundant uric acid crystals accompanying high serum phosphorus levels in a patient undergoing chemotherapy may indicate tumor lysis syndrome (TLS).

Fractional Excretion of Sodium and Urea

The urine-serum concentrations of sodium in relation to the urine-serum concentrations of creatinine (fractional excretion of sodium [FE_{Na}]) have been used to approximate kidney tubular function:

$$FE_{Na} = \frac{[U/S]_{Na}}{[U/S]_{Cr}} \times 100\%$$

where *U* is urine, *S* is serum, *Na* is s odium, and *Cr* is creatinine.

The basic premise is that renal tubular cells will reabsorb sodium in the prerenal setting, whereas tubules damaged by ATN will not.[41] FE_{Na} less than 1% is consistent with prerenal AKI, and FE_{Na} greater than 3% is typical of ATN. However, FE_{Na} may be less than 1% despite ATN in the setting of sepsis, hemoglobinuria or myoglobinuria, radiocontrast exposure, nonoliguria, heart failure, and advanced cirrhosis. Underlying CKD, diuretic use, recent intravenous fluid administration, glycosuria, bicarbonaturia, and salt-wasting disorders may be associated with elevated FE_{Na} despite the presence of prerenal AKI.[41] Therefore, FE_{Na} has significant limitations in the setting of hospital-acquired AKI, yet may be helpful in differentiating prerenal AKI from ATN in specific patient populations with oliguria.

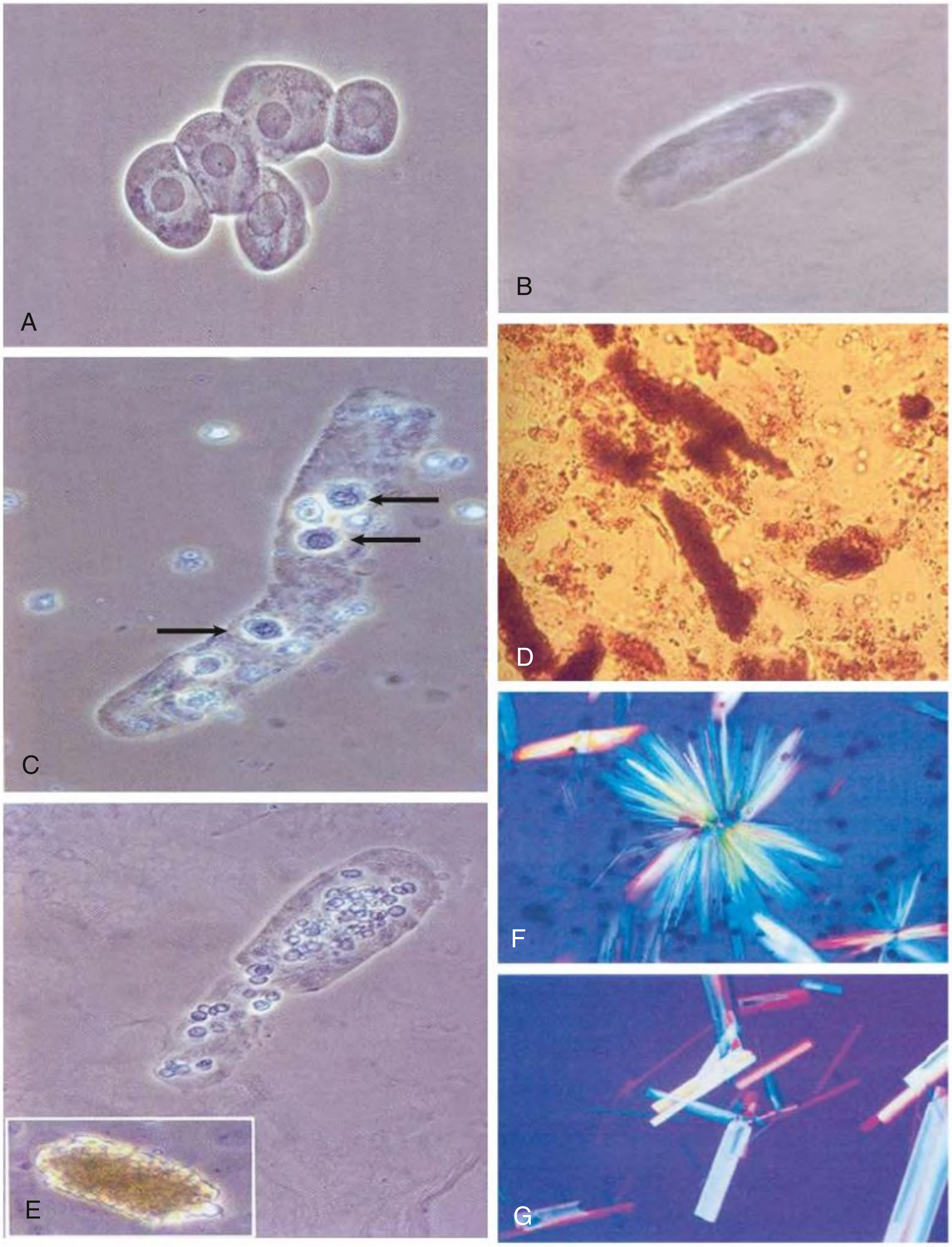

Fig. 72.3 Examples of Urinary Sediments Seen in Acute Kidney Injury (AKI). (A) Epithelial cell aggregate. (B) Hyaline cast as can be seen in prerenal AKI. (C) Epithelial cast as can be seen in early acute tubular necrosis (ATN; *arrows* indicate epithelial cells). (D) Muddy brown cast, typical of established ATN. (E) Erythrocyte cast as seen in glomerulonephritis and vasculitis. *Inset:* Hemoglobin cast. (F–G) Two forms of indinavir crystals.

Urea reabsorption, primarily occurring in proximal tubules, is less affected by loop and thiazide diuretics, and the fractional excretion of urea (FE_{Urea}) may be an alternative to FE_{Na} in patients receiving diuretics. The calculation of FE_{Urea} is identical to that of FE_{Na}, and values less than 35% favor prerenal AKI over ATN.

Laboratory Evaluation of Acute Kidney Injury in Systemic Illnesses

In the presence of systemic illness, additional laboratory evaluation may narrow the differential diagnosis of AKI. Serum complement levels of C3 and C4 are useful in differentiating causes of GN, particularly when a kidney biopsy is not feasible. Postinfectious GN, infective endocarditis, lupus nephritis, mixed cryoglobulinemia, and membranoproliferative GN are typically associated with low levels of serum C3 and/or C4 (see Chapter 22), whereas serum complement levels are normal in antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, anti–glomerular basement antibody disease, and immunoglobulin A (IgA) vasculitis (Henoch-Schönlein purpura). Despite activation of the complement system in thrombotic microangiopathy (see Chapter 30), serum C3 and C4 can be maintained at normal levels through increased hepatic production.[42] Alternatively, when liver synthesis is impaired, low

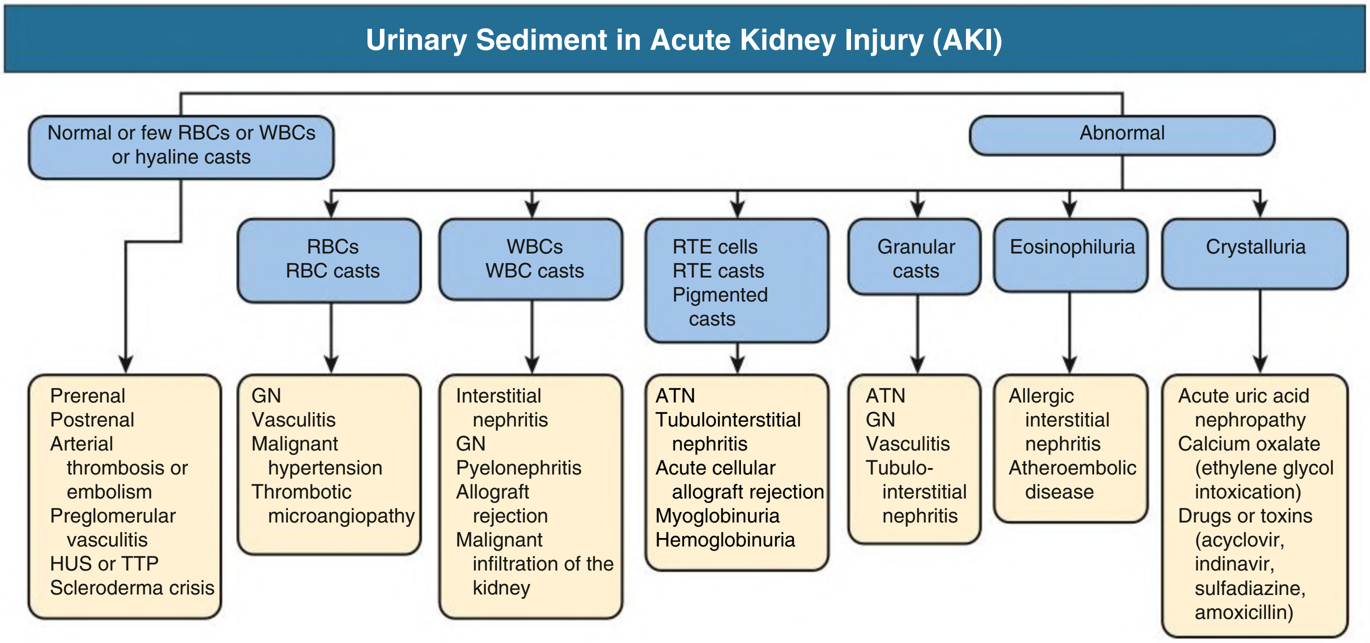

Fig. 72.4 Urinary sediment in acute kidney injury (AKI). *ATN,* Acute tubular necrosis; *GN,* glomerulonephritis; *HUS,* hemolytic uremic syndrome; *RBC,* red blood cell; *RTE,* renal tubular epithelial; *TTP,* thrombotic thrombocytopenic purpura; *WBC,* white blood cell.

serum levels of C3 and/or C4 are less reliable in differentiating the type of acute GN.

In the right clinical setting, laboratory studies may support a specific cause of AKI. For example, AKI with thrombocytopenia, schistocytes on peripheral blood smear, and an elevated blood lactate dehydrogenase level would be consistent with thrombotic microangiopathy from atypical hemolytic uremic syndrome or thrombotic thrombocytopenic purpura. However, these laboratory findings may also be present in sepsis with disseminated intravascular coagulation (DIC). AKI in sepsis with DIC typically results from ATN or antibiotic-related AIN and not thrombotic microangiopathy.

In the appropriate clinical setting, the following laboratory tests can help identify a cause: complete blood count with differential/platelet count, serum albumin, creatine kinase, uric acid, phosphorus, urine myoglobin, HIV antibody, hepatitis C antibody, cryoglobulins, rheumatoid factor, hepatitis B virus studies, antinuclear antibody, anti–double-stranded DNA antibody, ANCA, anti-GBM antibody, and serum FLC with ratio.

Imaging Studies

Kidney imaging may not be necessary if the explanation for AKI is apparent (e.g., prerenal AKI or ATN). However, when the diagnosis is uncertain, and especially if urinary obstruction or renal vascular occlusion are suspected, imaging is indicated.[27] Risk factors associated with hydronephrosis include history of prior hydronephrosis, current urinary tract infection, abdominal or pelvic cancer, and neurogenic bladder.[43] Point-of-care ultrasound is an emerging tool for assessing volume responsiveness in critically ill and ventilated patients, but it may also be a rapid and relatively low-cost option for excluding urinary obstruction in AKI (see Chapter 5).[44] A dedicated kidney ultrasound can identify urinary obstruction, polycystic kidney disease, and the size and number of kidneys. When Doppler flow is used, renal arteries and veins can be assessed. High-resolution, noncontrast computed tomographic imaging is the preferred test for the detection of urinary tract calculi. Radionuclide renography using radiolabeled mercaptoacetyltriglycine (MAG3) may be used to estimate the kidney plasma flow in a transplanted kidney with AKI but is increasingly replaced by Doppler ultrasound. Other radionuclide methods are less useful in AKI (e.g., tagged WBC scan for AIN). Magnetic resonance imaging without contrast is now recommended to evaluate renal arterial or venous thrombosis.[27]

Kidney Biopsy

Kidney biopsy is reserved for patients in whom prerenal and postrenal AKI have been excluded and the cause of intrinsic AKI remains unclear (see Fig. 72.2). Kidney biopsy is particularly useful when clinical assessment and laboratory investigations suggest diagnoses other than ischemic or nephrotoxic injury that may respond to disease-specific therapy (e.g., vasculitis, systemic lupus erythematosus, and AIN).

Electronic Health Record to Predict Acute Kidney Injury

Computer-based models have been used to develop AKI electronic alerts.[45,46] In one study, real-time alert of worsening of AKI by an AKI "sniffer" increased the timeliness of early therapeutic intervention.[47] However, in a large randomized controlled study, an electronic alert system for AKI in hospitalized patients did not demonstrate any improvements in clinical outcomes and rather increased the use of health care resources.[25] Further studies are needed to further define the utility of such AKI alerts.

ACUTE KIDNEY INJURY IN SPECIFIC SETTINGS

Acute Tubular Necrosis

ATN is a clinical syndrome of abrupt and sustained decline in GFR that is triggered by an acute ischemic or nephrotoxic event and develops within minutes to days after the insult.[27] Apart from kidney biopsy, no definitive test can diagnose ATN. The diagnosis is suggested by a history of recent hypotension, volume depletion, sepsis, or nephrotoxic exposure. Muddy brown, coarse, granular casts are present on urine microscopy in the majority of patients with ATN, particularly those who are oliguric.[27] Other laboratory findings consistent with the diagnosis of ATN are shown in Table 72.3.

The pathophysiology of ATN involves multiple pathways (see Chapter 70). Obstruction of flow within the tubule by cellular debris, disruptions of tubular cell polarity and cytoskeleton, loss of the lining epithelium with resulting backleak of glomerular filtrate into the kidney interstitium, and afferent arteriolar constriction all may play pathophysiologic roles in ATN. ATN has been described as a (mal)adaptive response of the kidney—"trading away" GFR for the preservation of medullary oxygenation and tubular integrity.[33]

ATN after ischemia/reperfusion injury is usually most severe within the outer medulla of the kidney (see Chapter 70).[48] The typical histologic features of human proximal tubular injury include vacuolation, loss of brush border, disruption of the epithelial cells lining the tubule, and presence of intratubular casts. Necrosis of tubular cells is typically focal and may be missed in a biopsy specimen because most biopsies contain cortex and the outer medulla is not sampled adequately. Apoptosis of tubular cells is present in kidney biopsy specimens from humans with ATN, and evidence of cellular regeneration is often seen, most commonly in areas of greatest tubular cell loss. Regenerative changes and fresh epithelial injury are often observed in the same biopsy specimen, suggesting that recurrent episodes of tubular ischemia continue to occur during the maintenance phase of ATN. The morphologic appearance of the common forms of nephrotoxin-induced ATN is similar to that of ischemic ATN, with loss of microvasculature and immune cell infiltration. Correlations of morphologic findings to functional endpoints have been difficult.

Acute Interstitial Nephritis

AIN is characterized by the presence of inflammatory infiltrates and edema within the interstitium, with an acute deterioration in GFR (see Chapter 64).[49] It is a relatively common cause of AKI, accounting for 15% to 27% of kidney biopsies performed because of AKI, yet AIN may be overlooked as a cause of AKI in settings in which ATN is common (e.g., sepsis). Drug use, in particular antimicrobials, proton pump inhibitors, and NSAIDs, is the most common cause of AIN (see Table 72.2).[49]

The diagnosis of AIN generally requires kidney biopsy. The clinical presentation of drug-induced AIN is variable, with a mean delay between drug exposure and the appearance of kidney manifestations of 10 days, although the latent period can be as short as 1 day with some antibiotics or as long as several months with NSAIDs.[49] In AIN the reported frequency of symptoms included arthralgias (45%), fever (36%), and skin rash (22%). In the same analysis, urinary findings included nonnephrotic proteinuria (93%), leukocyturia (82%), and microscopic hematuria (67%).[49] The classic clinical presentation of maculopapular rash, peripheral eosinophilia, and arthralgias may not be present in drug-induced AIN and is uncommon in infection-related or idiopathic AIN (see Table 72.2). Although the presence of urine eosinophils is not sensitive or specific for AIN, urinary levels of tumor necrosis factor α and interleukin-9 show promise as biomarkers of AIN.[50]

Acute Kidney Injury From Intratubular Obstruction

Several endogenous and exogenous molecules can precipitate in the tubular lumen to produce AKI, including uric acid, calcium phosphate, calcium oxalate, immunoglobulin FLCs, myoglobin, and medications (e.g., acyclovir, indinavir, methotrexate, sodium phosphate–containing cathartics, and sulfadiazine).[27] Medication-induced crystal nephropathy occurs more commonly when high doses of medication are combined with low tubular flow rates from volume contraction or underlying CKD.

Ethylene glycol ingestion from solvents and antifreeze may result in AKI from calcium oxalate crystal deposition. Patients with ethylene glycol poisoning typically have high anion gap metabolic acidosis because the ethylene glycol is metabolized into glycolic acid. The glycolic acid is converted to oxalic acid, which binds free calcium to form calcium oxalate crystals. Crystal deposition resulting in AKI typically manifests 48 to 72 hours after ingestion; oxalate crystals can be readily identified in the urine soon after exposure.

Intratubular obstruction from calcium phosphate and urate crystals contribute to AKI in TLS, which is characterized by a constellation of metabolic derangements caused by massive and abrupt release of intracellular components in the blood after rapid lysis of malignant cells. It is typically seen after initiation of cytotoxic therapy for hematologic malignancies with large tumor burden or cell counts.[51] Clinical features of TLS result from the effects of the metabolic derangements. Nucleic acids released during cell lysis are metabolized to hypoxanthine and then xanthine, which is converted by xanthine oxidase to uric acid. With significant tumor lysis, severe hyperuricemia and intratubular obstruction from urate crystals may result. Hyperkalemia may induce cardiac arrhythmias, weakness ("spaghetti legs"), and paresthesias. Hyperphosphatemia initially produces muscle cramps and lethargy but also can promote nausea, vomiting, diarrhea, and seizure. Hypocalcemia, primarily caused by phosphorus binding, causes similar symptoms with muscle cramps, tetany, cardiac arrhythmias, and seizures.[51] Hyperkalemia and hypocalcemia associated with AKI in a patient receiving cytotoxic chemotherapy may be the only initial clues to the syndrome because blood uric acid and phosphate levels are not routinely monitored. Urine microscopy can assist in the diagnosis by revealing the presence of many urate crystals.

Rhabdomyolysis

Rhabdomyolysis is characterized by the leakage of muscle-cell contents, including myoglobin, electrolytes, creatine kinase, aldolase, lactate dehydrogenase, nucleic acids, and aspartate aminotransferase into the circulation.[52] AKI, occurring primarily in severe rhabdomyolysis, results from renal vasoconstriction, proximal tubular cell injury from oxidant stress, and intranephronal obstruction. Both myoglobin, a heme pigment protein that contains iron in the ferrous (Fe^{2+}) state, and uric acid have less nephrotoxicity in alkaline urine. Intravascular volume contraction and acidic urine promote distal tubule obstruction from myoglobin and uric acid precipitation. A recent study showed that macrophages activated by platelets are able to form macrophage extracellular traps that are pathogenic in rhabdomyolysis-induced kidney injury.[53] See Table 72.4 for common causes of rhabdomyolysis.[52]

Patients with acute rhabdomyolysis often present with muscle pain and reddish-brown urine. The presence of pigmented granular casts and the lack of RBCs on urine microscopy coupled with a blood-positive urinary dipstick are important diagnostic clues for rhabdomyolysis-related AKI. However, the diagnosis must be confirmed by elevated serum creatine kinase levels and the presence of urinary myoglobin.[27] A weak correlation exists between creatine kinase values and the incidence of AKI. The risk for AKI is lower when creatine kinase levels are less than 20,000 U/L. Rhabdomyolysis may contribute to AKI with creatinine kinase levels as low as 5000 U/L when coexisting conditions such as sepsis, intravascular volume contraction, and acidosis are present.[52]

TABLE 72.4 **Causes of Rhabdomyolysis**[52]

Trauma[a]	Increased Exertion	Genetic Defects	Infections	Body Temperature Changes	Drug or Toxin Exposure	Metabolic and Electrolyte Disorders
Crush injury Prolonged immobilization Limb compression from loss of consciousness	Seizure Alcohol withdrawal Strenuous exercise	Disorders of glycolysis or gluconeogenesis Disorders of lipid metabolism Mitochondrial disorders	Influenza A and B Coxsackievirus Epstein-Barr virus Streptococcus pyogenes Pyomyositis *(Staphylococcus aureus)* *Clostridioides*	Heat stroke Neuroleptic malignant syndrome Hypothermia Malignant hyperthermia	Lipid-lowering drugs Alcohol[a] Quetiapine Cocaine[a] Heroin[a]	Hypokalemia Hypophosphatemia Hypocalcemia Nonketotic hyperosmotic conditions Diabetic ketoacidosis

[a]Most commonly associated with acute kidney injury.

Acute Kidney Injury in Myeloma

Monoclonal immunoglobulin FLCs are responsible for the majority of severe AKI in patients with myeloma. In such patients, monoclonal FLCs are overproduced, often with circulating levels hundreds of times higher than normal.[54] Unlike most endogenously produced proteins, monoclonal FLCs have a strong propensity to cause tubular damage (see Chapter 28).[54] Some κ monoclonal FLCs are cytotoxic and promote proximal tubular cell injury, with one mechanism resulting in a defect in sodium-coupled cotransport processes producing type II renal tubular acidosis, aminoaciduria, phosphaturia, and glycosuria (i.e., Fanconi syndrome).[55] A separate mechanism of FLC-mediated AKI is intratubular obstruction from precipitation of monoclonal FLCs in the distal nephron, that is, cast nephropathy.[27] Cast formation occurs under specific conditions mediated by the ionic composition of tubule fluid, tubule fluid flow rates, the concentration of Tamm-Horsfall glycoprotein and FLCs, the strength of the binding interaction between Tamm-Horsfall glycoprotein and FLC, and the presence of furosemide.[54] AKI attributed to cast nephropathy occurs in approximately one-third of patients with multiple myeloma and AKI. Other causes of AKI include extrarenal obstruction (e.g., nephrolithiasis, amyloid deposition in the ureters), hypercalcemia, hyperviscosity syndrome, and non–myeloma-related causes (e.g., AIN or contrast-associated AKI).[27,56]

Cast nephropathy should be considered, particularly in older patients with unexplained AKI. Predisposing conditions may include hypercalcemia, intravascular volume contraction, furosemide use, exposure to radiocontrast, and/or NSAID use. AKI may be the presenting event in patients with an undiagnosed plasma cell dyscrasia. Because FLC may not be detected on dipstick analysis, urinalysis in cast nephropathy will often show no or trace protein and the urinary sediment is typically bland. Assay of serum FLC levels is critical in assisting in the differential diagnosis of unexplained AKI (Chapter 68), but kidney biopsy may be needed to diagnose cast nephropathy. A low urinary albumin excretion (<10%) assessed with 24-hour urine electrophoresis was found useful in distinguishing cast nephropathy from ATN and less acute causes of kidney injury in multiple myeloma (e.g., amyloid light-chain [AL] amyloidosis and monoclonal immunoglobulin deposition disease).[57]

Contrast-Associated AKI

Historically, AKI occurring shortly after exposure to intravenous radiocontrast was termed *contrast-induced nephropathy*. However, iodinated contrast exposure may be one of multiple causes of AKI following a radiographic study or procedure. In a retrospective analysis with statistical adjustment, incidence rates of AKI were similar between contrast and noncontrast groups (5.5% vs. 5.6%, respectively).[58] The broader term, contrast-associated AKI, has gained favor in describing the pathophysiology of an AKI after contrast exposure.[56] The incidence of contrast-associated AKI is low (<1%) in patients with normal kidney function and no other risk factors for AKI.[27] Risk factors for contrast-associated AKI include CKD, diabetic nephropathy, advanced heart failure, states of reduced kidney perfusion, high total dose of contrast, and concomitant exposure to other nephrotoxins. Data from animal models suggest that both renal vasoconstriction from direct effects of the contrast media and toxic injury of tubular cells (likely from activation of osmolality-sensitive aldose reductase in the proximal tubule) are the main factors in the pathophysiology of contrast-associated AKI.[59] Emerging data suggest a role for contrast-induced uricosuria in the pathogenesis of contrast-associated AKI.[60] The kidney injury likely occurs within minutes of contrast exposure; however, the detection of AKI is typically delayed by 24 to 48 hours after contrast exposure. The timing of AKI in relation to contrast exposure and the exclusion of other causes of AKI generally suffice for diagnosis. Urinalysis and urine sediment findings are typically consistent with ATN. Kidney biopsy is generally not helpful in the setting of contrast-associated AKI because the expected findings of ATN are nonspecific, no specific treatment exists, and the kidney injury is typically short lived.

SELF-ASSESSMENT QUESTIONS

A 20-year-old man with Marfan syndrome is admitted to the hospital for elective aortic valve replacement and endovascular repair of a distal (type B) thoracic aortic dissection. On postoperative day 4, his urine output decreases to 420 mL over a 24-hour period and the urine appears dark in his Foley catheter collection bag. He requires mechanical ventilation and pressor support, with a mean arterial blood pressure of 70 mm Hg. There is no rash visible on examination. Available laboratory data are as follows.

Preoperation Blood		Postoperation Day 4 Blood		Postoperation Day 4 Blood	
Sodium	140 mEq/L	Sodium	141 mEq/L	Arterial Blood Gas	
Potassium	4.0 mEq/L	Potassium	6.1 mEq/L	pH	7.27
Chloride	108 mEq/L	Chloride	106 mEq/L	P_{CO_2}	31 mm Hg
Bicarbonate	24 mEq/L	Bicarbonate	14 mEq/L	P_{O_2}	97 mm Hg
BUN	12 mg/dL	BUN	24 mg/dL	HCO_3^-	14 mEq/L
Creatinine	0.6 mg/dL	Creatinine	4.4 mg/dL	Urinalysis	
Glucose	100 mg/dL	Glucose	98 mg/dL	Specific gravity	1.028
		Calcium	5 mg/dL	pH	6.0
		Phosphate	10.2 mg/dL	Protein	Negative
		Albumin	3.8 g/dL	Blood	Positive
		Osmolality	294 mOsm/kg	Ketones	Negative
		Lactate	Normal	Red blood cells	None

1. What is the most appropriate next diagnostic test?
   A. Serum creatine kinase level
   B. Kidney ultrasound with Doppler
   C. Kidney biopsy
   D. Serum antineutrophil cytoplasmic antibody
   E. Fractional excretion of urea
2. Which test result is most consistent with preserved kidney tubular function in a person who is volume contracted and has healthy kidneys?
   A. Urine osmolality of 300 mOsm/kg
   B. Fractional excretion of urea of 45%
   C. BUN of 34 mg/dL
   D. Spot urine sodium of 40 mEq/L
   E. None of the above
3. A 78-year-old man is brought to the emergency department by his family because of worsening confusion throughout the day. One week ago the patient fell and fractured his distal radius, necessitating splinting and pain medication. His medical history was pertinent with chronic kidney disease stage 3 (baseline serum creatinine 1.4 mg/dL), longstanding hypertension, chronic obstructive pulmonary disease, osteopenia, and benign prostatic hypertrophy. Current medications included lisinopril 20 mg/d, amlodipine 5 mg/d, hydrochlorothiazide 25 mg/d, doxazosin 4 mg nightly, oxycodone 5 mg every 8 hours as needed for arm pain, vitamin D 2000 U/d, and ipratropium bromide–albuterol inhaled 4 times per day. Arrival vital signs include blood pressure 176/92 mm Hg, pulse 114/min, respirations 18/min, weight 55 kg, and temperature 98.5°F. Physical examination revealed an agitated older man with flat neck veins, reduced air movement on chest auscultation, a large and tender abdomen, a right forearm cast, and trace lower extremity edema. No skin lesions were seen or palpated. Initial laboratory test results were as follows: sodium 138 mEq/L, potassium 5.2 mEq/L, chloride 110 mEq/L, bicarbonate 20 mEq/L, BUN 54 mg/dL, serum creatinine 2.6 mg/dL, leukocyte count 7500 cells/mL, hemoglobin 11.2 g/dL, platelet count 150,000 cells/µmol, alanine aminotransferase 28 U/L (normal 15–58 U/L), aspartate aminotransferase 36 U/L (normal 15–40 U/L), and prothrombin time 14 seconds. Urinalysis with microscopy: specific gravity 1.010, no blood or protein, and a bland sediment with no casts and few cells. What is the most appropriate next diagnostic test?
   A. Kidney ultrasound with Doppler
   B. Peripheral blood smear
   C. Kidney biopsy
   D. Postvoid bladder catheterization
   E. None of the above

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